infoblox

NETWORK OBSERVABILITY POWERED BY KENTIK

Kentik Deployment Guide

Prepared for Fabrikam Logistics

Global multi-cloud network observability

Customer	Fabrikam Logistics
Project	Global multi-cloud network observability
Prepared by	Infoblox Solutions Engineering
Date	2026-10-29
Kentik region	European Union (Frankfurt)
Scope	4 data center(s), 2 branch(es), AWS, Azure, Google Cloud, Oracle Cloud, Synthetic monitoring

This guide describes the Kentik deployment designed for this environment: what gets installed, which permissions and firewall rules are needed, and the steps to go live. Facts are taken from the Kentik Knowledge Base (kb.kentik.com) and each section names its sources.

Infoblox Confidential. Prepared for the named customer only.

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1. Deployment at a glance

Kentik is SaaS. The customer runs a small footprint where the network lives, and everything lands in the customer's Kentik account in the European Union (Frankfurt) region over encrypted, outbound connections.

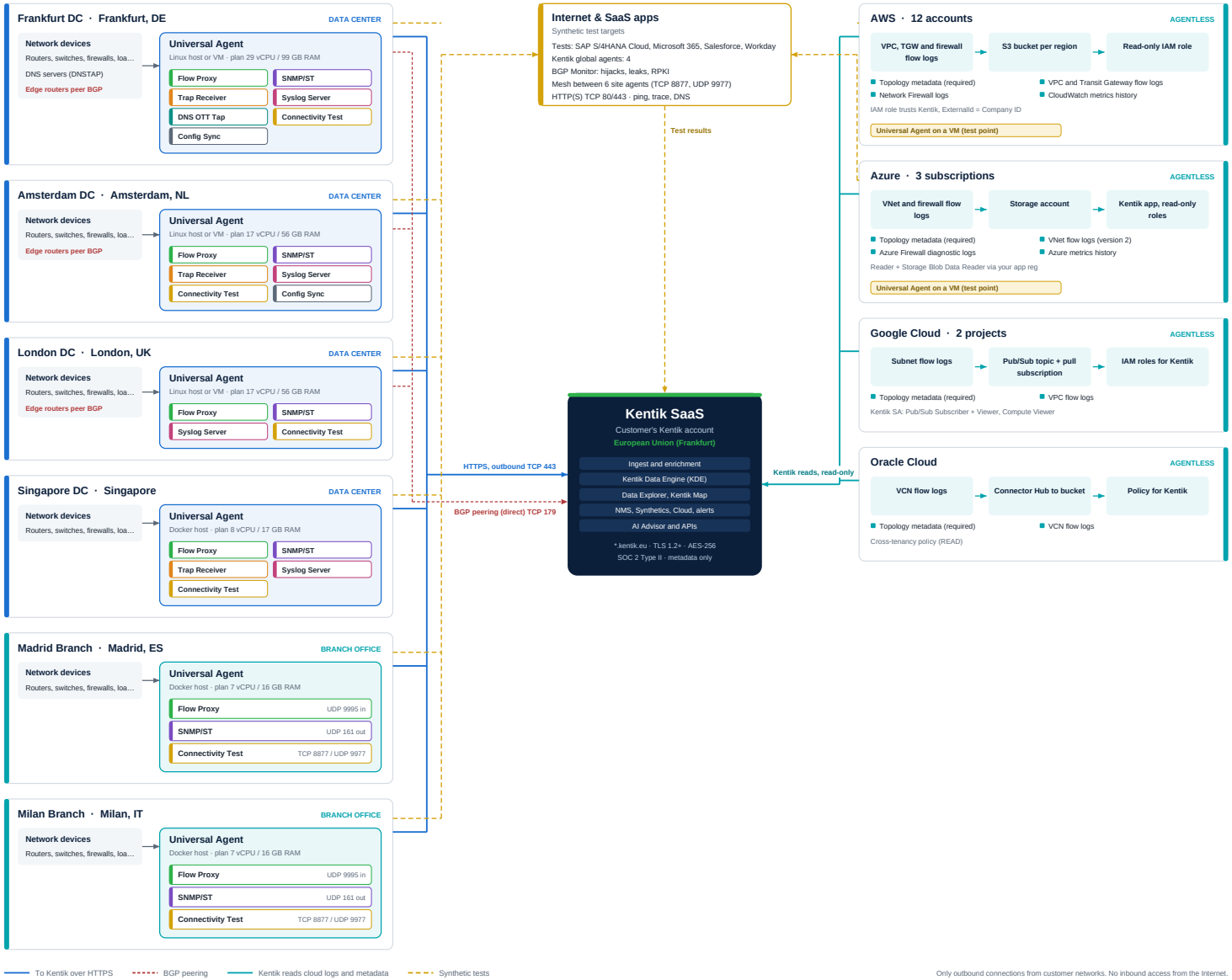
Where	Deployment	What Kentik collects	Footprint
Frankfurt DC (Frankfurt, DE)	Universal Agent on Linux	Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, DNS OTT Tap, Connectivity Test, Device Configuration Sync; edge routers peer BGP directly with Kentik	29 vCPU / 99 GB
Amsterdam DC (Amsterdam, NL)	Universal Agent on Linux	Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, Device Configuration Sync; edge routers peer BGP directly with Kentik	17 vCPU / 56 GB
London DC (London, UK)	Universal Agent on Linux	Flow Proxy Edge Processing, SNMP/ST, Syslog Server, Connectivity Test; edge routers peer BGP directly with Kentik	17 vCPU / 56 GB
Singapore DC (Singapore)	Universal Agent on Docker	Flow Proxy Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test; no BGP peering	8 vCPU / 17 GB
Madrid Branch (Madrid, ES)	Universal Agent on Docker	Flow Proxy Forwarding, SNMP/ST, Connectivity Test	7 vCPU / 16 GB
Milan Branch (Milan, IT)	Universal Agent on Docker	Flow Proxy Forwarding, SNMP/ST, Connectivity Test	7 vCPU / 16 GB
Amazon Web Services	Agentless (read-only access)	Topology metadata (required), VPC and Transit Gateway flow logs, Network Firewall logs, CloudWatch metrics history	Test agent VM
Microsoft Azure	Agentless (read-only access)	Topology metadata (required), VNet flow logs (version 2), Azure Firewall diagnostic logs, Azure metrics history	Test agent VM
Google Cloud	Agentless (read-only access)	Topology metadata (required), VPC flow logs	No compute
Oracle Cloud Infrastructure	Agentless (read-only access)	Topology metadata (required), VCN flow logs	No compute

Three patterns cover this design

- **On-prem: a Universal Agent per data center.** A single agent with only the capabilities each site needs toggled on in the portal. Devices point at it, and it talks to Kentik over outbound HTTPS. The standalone kproxy agent is not used; the Universal Agent Flow Proxy replaces it.
- **Public cloud: agentless.** Flow logs go to storage in the customer's own account, and Kentik gets read-only access to the logs, topology metadata and metrics.
- **Internet and SaaS: synthetic tests.** Kentik's global agents and the customer's own agents test SaaS apps, websites, DNS and paths between sites and clouds.

Source: Kentik KB, Universal Agents; Public Clouds; Synthetics Overview

2. Topology


Frankfurt DC · Frankfurt, DE
DATA CENTER
Network devices
Routers, switches, firewalls, loa…
DNS servers (DNSTAP)
Edge routers peer BGP
Universal Agent
Linux host or VM · plan 29 vCPU / 99 GB RAM
Flow Proxy
SNMP/ST
Trap Receiver
Syslog Server
DNS OTT Tap
Connectivity Test
Config Sync
Amsterdam DC · Amsterdam, NL
DATA CENTER
Network devices
Routers, switches, firewalls, loa…
Edge routers peer BGP
Universal Agent
Linux host or VM · plan 17 vCPU / 56 GB RAM
Flow Proxy
SNMP/ST
Trap Receiver
Syslog Server
Connectivity Test
Config Sync
London DC · London, UK
DATA CENTER
Network devices
Routers, switches, firewalls, loa…
Edge routers peer BGP
Universal Agent
Linux host or VM · plan 17 vCPU / 56 GB RAM
Flow Proxy
SNMP/ST
Syslog Server
Connectivity Test
Singapore DC · Singapore
DATA CENTER
Network devices
Routers, switches, firewalls, loa…
Universal Agent
Docker host · plan 8 vCPU / 17 GB RAM
Flow Proxy
SNMP/ST
Trap Receiver
Syslog Server
Connectivity Test
Madrid Branch · Madrid, ES
BRANCH OFFICE
Network devices
Routers, switches, firewalls, loa…
Universal Agent
Docker host · plan 7 vCPU / 16 GB RAM
Flow Proxy
UDP 9995 in
SNMP/ST
UDP 161 out
Connectivity Test
TCP 8877 / UDP 9977
Milan Branch · Milan, IT
BRANCH OFFICE
Network devices
Routers, switches, firewalls, loa…
Universal Agent
Docker host · plan 7 vCPU / 16 GB RAM
Flow Proxy
UDP 9995 in
SNMP/ST
UDP 161 out
Connectivity Test
TCP 8877 / UDP 9977
Internet & SaaS apps
Synthetic test targets
Tests: SAP S/4HANA Cloud, Microsoft 365, Salesforce, Workday
Kentik global agents: 4
BGP Monitor: hijacks, leaks, RPKI
Mesh between 6 site agents (TCP 8877, UDP 9977)
HTTP(S) TCP 80/443 · ping, trace, DNS
Test results
Kentik SaaS
Customer's Kentik account
European Union (Frankfurt)
Ingest and enrichment
Kentik Data Engine (KDE)
Data Explorer, Kentik Map
NMS, Synthetics, Cloud, alerts
AI Advisor and APIs
*.kentik.eu · TLS 1.2+ · AES-256
SOC 2 Type II · metadata only
HTTPS, outbound TCP 443
BGP peering (direct) TCP 179
Kentik reads, read-only
AWS · 12 accounts
AGENTLESS
VPC, TGW and firewall flow logs
S3 bucket per region
Read-only IAM role
Topology metadata (required)
VPC and Transit Gateway flow logs
Network Firewall logs
CloudWatch metrics history
IAM role trusts Kentik, ExternalId = Company ID
Universal Agent on a VM (test point)
Azure · 3 subscriptions
AGENTLESS
VNet and firewall flow logs
Storage account
Kentik app, read-only roles
Topology metadata (required)
VNet flow logs (version 2)
Azure Firewall diagnostic logs
Azure metrics history
Reader + Storage Blob Data Reader via your app reg
Universal Agent on a VM (test point)
Google Cloud · 2 projects
AGENTLESS
Subnet flow logs
Pub/Sub topic + pull subscription
IAM roles for Kentik
Topology metadata (required)
VPC flow logs
Kentik SA: Pub/Sub Subscriber + Viewer, Compute Viewer
Oracle Cloud
AGENTLESS
VCN flow logs
Connector Hub to bucket
Policy for Kentik
Topology metadata (required)
VCN flow logs
Cross-tenancy policy (READ)
To Kentik over HTTPS
BGP peering
Kentik reads cloud logs and metadata
Synthetic tests
Only outbound connections from customer networks. No inbound access from the Internet.

3. Roles and permissions

Who on the customer side does what, and the access each task needs. Kentik only ever receives read-only access.

Role	Task	Access needed
Kentik administrator	Deploy and authorize Universal Agents, enable capabilities, add devices, create cloud exports and tests	Kentik admin (or super admin) user in the customer's Kentik account
Server or platform team	Provide 6 agent host(s) and run the install command	root or sudo on a supported Linux host, or Docker
Network team	Point flow, traps and syslog at the agents; create SNMP and SSH credentials; configure BGP peering	Device configuration rights; SNMP v2c/v3 read-only credentials; read-only SSH account
Security / firewall team	Open the rules in section 4	Firewall and proxy change rights
DNS team	Stream DNSTAP from DNS servers (for example Infoblox NIOS) to the agent	DNS appliance administration
AWS IAM administrator	AWS: enable flow logs and grant Kentik access	IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket
Entra ID Application Administrator and subscription Owner	Azure: enable flow logs and grant Kentik access	Reader on subscriptions or resource groups; Storage Blob Data Reader on the log storage account
GCP project or folder IAM administrator	Google Cloud: enable flow logs and grant Kentik access	kentik-vpc-flow@kentik-vpc-flow.iam.gserviceaccount.com gets Pub/Sub Subscriber + Viewer on the subscription and Compute Viewer on the project or folder
OCI tenancy administrator	Oracle Cloud: enable flow logs and grant Kentik access	Cross-tenancy policy generated by the Kentik portal (recommended), or a user and group holding Kentik's API key; READ on the bucket and network resources

Access granted to Kentik

Scope	Access
AWS	IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket
Azure	Reader on subscriptions or resource groups; Storage Blob Data Reader on the log storage account
Google Cloud	kentik-vpc-flow@kentik-vpc-flow.iam.gserviceaccount.com gets Pub/Sub Subscriber + Viewer on the subscription and Compute Viewer on the project or folder
Oracle Cloud	Cross-tenancy policy generated by the Kentik portal (recommended), or a user and group holding Kentik's API key; READ on the bucket and network resources
Network devices	SNMP read-only (polled from the customer's own agent, never from the Internet)
Network devices	Read-only SSH from the agent; command block-list, never configure or commit

Source: Kentik KB, UA Deployment; NMS Setup; SSH Command Access for Kentik NMS; Kentik for AWS; Kentik for Azure; Kentik for GCP; Kentik for OCI

4. Network and firewall requirements

All Universal Agent connections are outbound. Nothing needs to reach into the customer network from the Internet.

Outbound allowlist (every agent)

Destination	Port	Notes
flow.kentik.eu	TCP 443	TLS 1.2+
grpc.api.kentik.eu	TCP 443	TLS 1.2+
api.kentik.eu	TCP 443	TLS 1.2+
portal.kentik.eu	TCP 443	TLS 1.2+
storage.googleapis.com	TCP 443	TLS 1.2+
whoami.kentiklabs.com	TCP 443	Optional
Kentik flow ingest	UDP 40010	Optional Flow Proxy base port

- SSL-inspecting firewalls and proxies must allow gRPC over port 443.
- UDP 40010 to Kentik is an optional Flow Proxy base port.
- A web proxy works: set https_proxy before running the install (Linux) or pass it with --env (Docker).

Frankfurt DC · Frankfurt, DE

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.eu, grpc.api.kentik.eu, api.kentik.eu, portal.kentik.eu, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Network devices	Universal Agent	UDP 162	SNMP traps
Network devices: routers, switches, firewalls, load balancers	Universal Agent	UDP and TCP 514	Syslog
DNS servers (for example Infoblox NIOS)	Universal Agent	DNSTAP port you choose	DNSTAP stream, or a SPAN or mirror port
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests

From	To	Port	Purpose
Universal Agent	Device management IPs	TCP 22	SSH config backup and read-only show commands
Edge routers	Kentik ingest peering IPs (shown in device settings)	TCP 179	BGP peering (direct, telemetry only)

Amsterdam DC · Amsterdam, NL

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.eu, grpc.api.kentik.eu, api.kentik.eu, portal.kentik.eu, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Network devices	Universal Agent	UDP 162	SNMP traps
Network devices: routers, switches, firewalls, load balancers	Universal Agent	UDP and TCP 514	Syslog
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Universal Agent	Device management IPs	TCP 22	SSH config backup and read-only show commands
Edge routers	Kentik ingest peering IPs (shown in device settings)	TCP 179	BGP peering (direct, telemetry only)

London DC · London, UK

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.eu, grpc.api.kentik.eu, api.kentik.eu, portal.kentik.eu, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)

From	To	Port	Purpose
Network devices: routers, switches, firewalls, load balancers	Universal Agent	UDP and TCP 514	Syslog
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests
Edge routers	Kentik ingest peering IPs (shown in device settings)	TCP 179	BGP peering (direct, telemetry only)

Singapore DC · Singapore

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.eu, grpc.api.kentik.eu, api.kentik.eu, portal.kentik.eu, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Network devices	Universal Agent	UDP 162	SNMP traps
Network devices: routers, switches, firewalls, load balancers	Universal Agent	UDP and TCP 514	Syslog
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests

Madrid Branch · Madrid, ES

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.eu, grpc.api.kentik.eu, api.kentik.eu, portal.kentik.eu, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)

From	To	Port	Purpose
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests

Milan Branch · Milan, IT

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.eu, grpc.api.kentik.eu, api.kentik.eu, portal.kentik.eu, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests

Cloud test agents

A Universal Agent on a VM in a cloud network uses the same outbound TCP 443 rules. Allow TCP 8877 and UDP 9977 between it and the other agents in security groups or NSGs.

Azure storage firewall

Only if the log storage account firewall is restricted: allow 20.69.189.228 and 20.69.185.115 (WestUS2: 51.8.214.254 and 172.171.46.16).

Source: Kentik KB, Universal Agents; UA Deployment; Test Types Reference; Azure Prerequisites & Roles

5. Universal Agent deployment

Install one Universal Agent per site, then switch capabilities on in the portal. The agent updates itself and reports its own health.

Host requirements

- Supported OS: Debian 11, 12, 13; Ubuntu 20.04, 22.04, 24.04; Oracle Linux 9; CentOS Stream 8. CentOS 7 and Enterprise Linux before 8 are not supported.
- Or any Docker host that can run the generated docker run command.
- Size for the sum of the capabilities on the host, plus 20 to 30% headroom for bursty flow, syslog and traps. The plans below use 30%.
- Disk sizing is not published in the KB; confirm with Kentik.

> **Rolling out many agents?** Automated rollout (optional): create a provisioning token with POST https://grpc.api.kentik.eu/kagent/v202401/provisioning-tokens (name, maxUsageCount, requiresApproval, config.siteId, optional expiresAt), export K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN on the host, then authorize with PUT .../agents/authorize/<install-id>. A token created with requiresApproval false skips authorization. Revoke the token when all agents have registered.

Source: Kentik KB, Universal Agents; UA Deployment

5.1 Frankfurt DC · Frankfurt, DE

Data center. Install method: **Linux package**. Capabilities: **Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, DNS OTT Tap, Connectivity Test, Device Configuration Sync**.

Capability	Planned load	CPU / RAM
Flow Proxy Edge Processing	30,000 flows/s (scaled from the KB figure: confirm with Kentik)	8 cores / 32 GB x 2
SNMP/ST	400 devices	2 cores / 8 GB
Syslog Server	4,000 events/s	1 core / 1 GB
DNS OTT Tap	80,000 lookups/s	1 core / 1 GB
Connectivity Test	per agent	2 cores / 2 GB
SNMP Trap Receiver	event driven	Not published
Device Configuration Sync	per fetch interval	Not published
Total from KB sizing		22 cores / 76 GB
Recommended host (+30%)		29 vCPU / 99 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.

4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Frankfurt DC), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, DNS OTT Tap, Connectivity Test, Device Configuration Sync) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

BGP peering from the edge routers

- In Kentik: device settings > BGP > Peer with device. Kentik shows its ingest peering IPv4 and IPv6 addresses next to the fields.
- Kentik peers as an iBGP route-reflector client using your ASN. 4-byte ASN support is required; an MD5 password is optional.
- Filter the default route toward Kentik so it does not override the destination AS for unmatched flows.
- BGP here is telemetry only; customer routing is unaffected. Peering also enables RTBH and Flowspec mitigation.
- Sites that use the BGP Proxy capability peer with their Universal Agent instead (see BGP Proxy below).

5.2 Amsterdam DC · Amsterdam, NL

Data center. Install method: **Linux package**. Capabilities: **Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, Device Configuration Sync**.

Capability	Planned load	CPU / RAM
Flow Proxy Edge Processing	20,000 flows/s	8 cores / 32 GB
SNMP/ST	250 devices	2 cores / 8 GB
Syslog Server	2,000 events/s	1 core / 1 GB
Connectivity Test	per agent	2 cores / 2 GB
SNMP Trap Receiver	event driven	Not published
Device Configuration Sync	per fetch interval	Not published
Total from KB sizing		13 cores / 43 GB
Recommended host (+30%)		17 vCPU / 56 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Amsterdam DC), then click Continue.

6. Toggle on the capabilities for this site (Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, Device Configuration Sync) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

BGP peering from the edge routers

- In Kentik: device settings > BGP > Peer with device. Kentik shows its ingest peering IPv4 and IPv6 addresses next to the fields.
- Kentik peers as an iBGP route-reflector client using your ASN. 4-byte ASN support is required; an MD5 password is optional.
- Filter the default route toward Kentik so it does not override the destination AS for unmatched flows.
- BGP here is telemetry only; customer routing is unaffected. Peering also enables RTBH and Flowspec mitigation.
- Sites that use the BGP Proxy capability peer with their Universal Agent instead (see BGP Proxy below).

5.3 London DC · London, UK

Data center. Install method: **Linux package**. Capabilities: **Flow Proxy Edge Processing, SNMP/ST, Syslog Server, Connectivity Test**.

Capability	Planned load	CPU / RAM
Flow Proxy Edge Processing	12,000 flows/s	8 cores / 32 GB
SNMP/ST	150 devices	2 cores / 8 GB
Syslog Server	1,500 events/s	1 core / 1 GB
Connectivity Test	per agent	2 cores / 2 GB
Total from KB sizing		13 cores / 43 GB
Recommended host (+30%)		17 vCPU / 56 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (London DC), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy Edge Processing, SNMP/ST, Syslog Server, Connectivity Test) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

> Behind a web proxy run export https_proxy=http://<proxy>:<port> before the install command.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
export https_proxy=http://<proxy>:<port>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

BGP peering from the edge routers

- In Kentik: device settings > BGP > Peer with device. Kentik shows its ingest peering IPv4 and IPv6 addresses next to the fields.
- Kentik peers as an iBGP route-reflector client using your ASN. 4-byte ASN support is required; an MD5 password is optional.
- Filter the default route toward Kentik so it does not override the destination AS for unmatched flows.
- BGP here is telemetry only; customer routing is unaffected. Peering also enables RTBH and Flowspec mitigation.
- Sites that use the BGP Proxy capability peer with their Universal Agent instead (see BGP Proxy below).

5.4 Singapore DC · Singapore

Data center. Install method: **Docker**. Capabilities: **Flow Proxy Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test**.

Capability	Planned load	CPU / RAM
Flow Proxy Forwarding	8,000 flows/s	1 core / 2 GB
SNMP/ST	90 devices	2 cores / 8 GB
Syslog Server	800 events/s	1 core / 1 GB
Connectivity Test	per agent	2 cores / 2 GB
SNMP Trap Receiver	event driven	Not published
Total from KB sizing		6 cores / 13 GB
Recommended host (+30%)		8 vCPU / 17 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Docker. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the container ID). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Singapore DC), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
# run the docker command copied from Settings > Universal Agents > Deploy Agent
# it carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN
```

5.5 Madrid Branch · Madrid, ES

Branch office. Install method: **Docker**. Capabilities: **Flow Proxy Forwarding, SNMP/ST, Connectivity Test**.

Capability	Planned load	CPU / RAM
Flow Proxy Forwarding	1,500 flows/s	1 core / 2 GB
SNMP/ST	15 devices	2 cores / 8 GB
Connectivity Test	per agent	2 cores / 2 GB
Total from KB sizing		5 cores / 12 GB
Recommended host (+30%)		7 vCPU / 16 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Docker. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the container ID). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Madrid Branch), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy Forwarding, SNMP/ST, Connectivity Test) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
# run the docker command copied from Settings > Universal Agents > Deploy Agent
# it carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN
```

5.6 Milan Branch · Milan, IT

Branch office. Install method: **Docker**. Capabilities: **Flow Proxy Forwarding, SNMP/ST, Connectivity Test**.

Capability	Planned load	CPU / RAM
Flow Proxy Forwarding	1,500 flows/s	1 core / 2 GB
SNMP/ST	15 devices	2 cores / 8 GB
Connectivity Test	per agent	2 cores / 2 GB
Total from KB sizing		5 cores / 12 GB
Recommended host (+30%)		7 vCPU / 16 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Docker. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.

4. The agent appears under Select an Agent (host name is the container ID). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Milan Branch), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy Forwarding, SNMP/ST, Connectivity Test) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
# run the docker command copied from Settings > Universal Agents > Deploy Agent
# it carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN
```

Source: Kentik KB, UA Deployment; UA Portal Management; Device Settings

6. Capability configuration

How to configure each capability used in this design, on the agent and on the devices that feed it.

Flow Proxy

Receives NetFlow, sFlow and IPFIX from routers, switches and firewalls and sends it to Kentik over TLS. Forwarding sends the raw flow to Kentik SaaS, which decodes it, so it needs far less CPU and RAM. Edge Processing decodes and enriches the flow on the agent before sending it, so it needs a much larger host.

Enabled at: Frankfurt DC (Edge Processing), Amsterdam DC (Edge Processing), London DC (Edge Processing), Singapore DC (Forwarding), Madrid Branch (Forwarding), Milan Branch (Forwarding) · **License:** FlowPak

- Toggle on Flow Proxy on the agent in the mode chosen for this site (Forwarding or Edge Processing). Settings: Host 0.0.0.0, Port 9995 (UDP), optional sFlow agent address override, and Site.
- On each device, export flow to the agent's IP on UDP 9995. Sample ingress on every interface; never ingress and egress on the same interface.
- The sample rate on the device must match the rate set for that device in Kentik.
- Vendor configuration snippets for flow, SNMP and BGP: github.com/kentik/config-snippets.
- Sanity check after go-live: flow volume should land within about 20% of SNMP interface traffic.

Source: Kentik KB, Universal Agents; Router Configuration; Flow Overview; Migration Scenarios (kproxy to UA)

SNMP/ST

Discovers devices and collects health, interface and hardware metrics by SNMP polling or gNMI streaming telemetry. It also enriches flow with interface names and speeds.

Enabled at: Frankfurt DC, Amsterdam DC, London DC, Singapore DC, Madrid Branch, Milan Branch · **License:** FlowPak (flow enrichment) or NMS / DevicePak (full monitoring)

- Run the NMS Setup Wizard: pick this agent, enter IP ranges and SNMP v1, v2c or v3 credentials, discover, then add devices.
- Per device, on the SNMP tab choose "Flow Enrichment via Universal Agent" or "Full Monitoring via Universal Agent".
- SNMP v3 options: auth None, MD5 or SHA; privacy None, DES or AES.
- Default polling: 1 minute for status, CPU, memory and interfaces; 5 minutes for everything else.
- Streaming telemetry (optional): enable "Dial-In from Universal Agent" (gNMI over gRPC, 30 second default).

Source: Kentik KB, Universal Agents; NMS Setup; Device Settings

SNMP Trap Receiver

Captures SNMP traps the moment a device sends them, for example an interface or BGP session going down.

Enabled at: Frankfurt DC, Amsterdam DC, Singapore DC · **License:** NMS / DevicePak

- Enable SNMP Trap Receiver on the agent.
- Point device trap targets at the agent's IP on UDP 162.
- Traps must come from a device licensed for NMS. Query them in Data Explorer with the metric "SNMP Traps: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

Syslog Server

Collects log messages from network devices (routers, switches, firewalls, load balancers) so they can be searched, alerted on and lined up with traffic.

Enabled at: Frankfurt DC, Amsterdam DC, London DC, Singapore DC · **License:** NMS / DevicePak

- Enable Syslog Server on the agent.
- Point network device syslog at the agent's IP on UDP or TCP 514.
- If a syslog forwarder sits in between, it must keep the original device IP. Changing the port currently needs Kentik support.
- Query in Data Explorer with the metric "Syslog: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

DNS OTT Tap

Turns DNS lookups into records Kentik can join with traffic, for OTT service tracking and DNS troubleshooting.

Enabled at: Frankfurt DC · **License:** Confirm with the Kentik account team

- Enable DNS OTT Tap on the agent.
- Preferred mode: DNSTAP receiver, for DNS appliances such as Infoblox. Other modes: promiscuous capture of a SPAN or mirror port, or directly on a BIND9 or Linux resolver.
- On the DNS servers, stream DNSTAP to the agent's IP and the port you configure.

Source: Kentik KB, Universal Agents

Connectivity Test

Makes this agent a private test point. It measures latency, loss and jitter to other sites, clouds, SaaS apps and DNS servers, and gives AI Advisor on-demand ping and traceroute.

Enabled at: Frankfurt DC, Amsterdam DC, London DC, Singapore DC, Madrid Branch, Milan Branch · **License:** Synthetics test credits

- Toggle on Connectivity Test on the agent.
- Open TCP 8877 and UDP 9977 between every pair of agents that will test each other.
- Create tests in Synthetics: agent-to-agent, network mesh, server IP or hostname, DNS server monitor and HTTP(S) or API tests.
- Confirm which continuous test types run on the Universal Agent today with the Kentik account team.

Source: Kentik KB, Universal Agents; Synthetics Overview; Test Types Reference; Test Credits

Device Configuration Sync

Backs up configs, shows what changed, and runs read-only show commands during an AI Advisor investigation.

Enabled at: Frankfurt DC, Amsterdam DC · **License:** NMS / DevicePak

- Create a read-only SSH account on each device (TACACS+ or RADIUS logs will show the agent as the source).
- In device settings, SSH tab: credential, collection agent, platform autodetect, config sync and fetch interval (5 minutes recommended for high-visibility environments).
- Supported platforms: Juniper Junos, Cisco NX-OS, Arista EOS and Cisco IOS-XE.
- Read-only by design: a command block-list, never configure or commit.

Source: Kentik KB, SSH Command Access for Kentik NMS; Device Settings

7. Public cloud setup

No agents are needed in public clouds. Each cloud writes flow logs to storage in the customer's own account, and Kentik reads them with read-only access. Kentik recommends a metadata-only export first to prove access before turning on high-volume logs.

Amazon Web Services

No agent. Flow logs are written to S3 buckets the customer owns. Kentik's collector, running in Kentik's own AWS account, reads them inside AWS and sends them on to Kentik SaaS, so the customer pays no egress for the logs. Metadata API calls come from Kentik SaaS through a read-only IAM role, protected by an ExternalId.

Collecting: Topology metadata (required), VPC and Transit Gateway flow logs, Network Firewall logs, CloudWatch metrics history. Scope: 12 account(s)/subscription(s)/project(s). Regions: eu-central-1, eu-west-1.

Steps

1. Start with a metadata-only export to prove access before turning on high-volume logs.
2. Create one S3 bucket per region for flow logs. Kentik treats each bucket as one device, so hundreds of VPCs into one global bucket risks S3 throttling.
3. Encryption: SSE-S3 is recommended. SSE-KMS needs the KMS statement. SSE-C is not supported.
4. Turn on flow logs for VPCs, Transit Gateways and Network Firewalls: filter All, destination S3, custom format with every v2 to v5 field.
5. Create the IAM role with the trust policy and permissions below.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > AWS Cloud. Select features, enter the Role ARN > Verify Role, Region > Verify Region, bucket name and prefix > Verify S3 Bucket.
7. Pick sampling (Legacy, a rate from 2 to 2000, or Unsampled), name the export, choose the billing plan and Save.
8. Optional: tick "Help me configure my provider via Terraform" for a main.tf that creates the role, bucket and flow logs.

How data moves

- Flow logs: Kentik's collector runs in Kentik's AWS account (834693425129). It assumes your role and reads new log files from your S3 buckets inside AWS, then sends them to Kentik SaaS, so you don't pay egress charges for the logs.
- Metadata: Kentik SaaS calls the AWS describe APIs (EC2, Direct Connect, ELB, Network Firewall and so on) through the same read-only role, using sts:AssumeRole with your ExternalId.

IAM role trust policy (primary account)

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": { "AWS": "arn:aws:iam::834693425129:root" },
    "Action": "sts:AssumeRole",
    "Condition": {
      "StringEquals": { "sts:ExternalId": "<Kentik Company ID>" },
      "ArnEquals": { "aws:PrincipalArn": "arn:aws:iam::834693425129:role/eks-ingest-node" }
    }
  }]
}
```

Nested accounts: primary and secondary roles

Kentik's account assumes the primary role in account A. The primary role then assumes a secondary role in each other account (B, C and so on) to read its metadata.

1. Pick the primary (hub) account, for example the account that holds the log buckets. Kentik assumes only the primary role.
2. In the primary account, create the primary role with the trust policy above (Kentik's account, ExternalId = Company ID).
3. Give the primary role a policy with sts:AssumeRole on the secondary roles ("AssumeSecondaryRoles") and organizations:ListAccounts.
4. In each secondary account, create a secondary role that trusts the primary account (arn:aws:iam::<primary account ID>:root, sts:AssumeRole, no ExternalId) and attach the read-only metadata permissions below. Use the same role name in every account.
5. In the Kentik export, tick "This is an AWS organization role", then enter the secondary account IDs, their regions and the secondary role name.
6. Flow logs from the secondary accounts can be sent to a bucket in the primary account (a central log archive), so one export reads them all.

Secondary role trust policy

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": { "AWS": "arn:aws:iam::<primary account ID>:root" },
    "Action": "sts:AssumeRole",
    "Condition": {}
  }]
}
```

IAM permissions policy

Service	Actions
EC2	ec2:Describe*, ec2:Search*, ec2:GetManagedPrefixListEntries
Direct Connect	directconnect:List*, directconnect:Describe*
ELB	elasticloadbalancing:Describe*
IAM	iam:ListAccountAliases
Network Firewall	network-firewall:Describe*, network-firewall:List*
Cloud WAN	networkmanager:ListCoreNetworks, GetCoreNetwork, GetCoreNetworkPolicy, ListAttachments, GetNetworkRoutes
CloudWatch	cloudwatch:ListMetrics, GetMetricStatistics, GetMetricData, Describe*
Organizations	organizations:ListAccounts
S3 (log bucket)	s3:Get*, s3:List* on arn:aws:s3:::<bucket> and arn:aws:s3:::<bucket>/*
KMS (SSE-KMS only)	kms:Decrypt, kms:DescribeKey on the bucket key

Export status values: Start, Pending, Error, OK, Disabled or Halt. Licensing: CloudPak, measured in flows per second.

Test agent in AWS: install a Universal Agent on a small VM with the Synthetics capability, using the same steps as on-prem. It needs outbound TCP 443 only, plus TCP 8877 and UDP 9977 to the other agents.

Source: Kentik KB, Kentik for AWS; Metadata Configuration (AWS); Flow/Firewall Log Collection Configuration (AWS); Kentik Export Configuration for AWS; Public Clouds

Microsoft Azure

No agent. Network Watcher writes VNet flow logs to a storage account the customer owns, and Kentik reads it through an app approved in Entra ID.

Collecting: Topology metadata (required), VNet flow logs (version 2), Azure Firewall diagnostic logs, Azure metrics history. Scope: 3 account(s)/subscription(s)/project(s).

Steps

1. Create a dedicated storage account for the logs.
2. If the storage account firewall is restricted, allow 20.69.189.228 and 20.69.185.115 (WestUS2 uses 51.8.214.254 and 172.171.46.16).
3. In Network Watcher turn on VNet flow logs (version 2) to that storage account.
4. Send Azure Firewall diagnostic logs to the same storage account with the azfwapplicationrule and azfwnetworkrule categories.
5. Authorize Kentik: create a single-tenant app registration with your own secret or certificate, and enter it in Kentik as a credential with the client and tenant IDs.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > Azure Cloud. Pick features, API access method, location and subscription ID, Validate Service Principal, resource group > Verify Access.
7. Enter the storage account name, sampling (2 to 2000 or Unsampled), optional enrichment scope and billing plan, then Save.
8. Optional: the export wizard can generate a PowerShell script that does most of the Azure side (VNet flow logs only).

Roles for the Kentik service principal

Role	Scope	Why
Reader	Subscription or resource group holding the logs	Read resources
Reader	Additional subscriptions or resource groups	Enrichment scope
Storage Blob Data Reader (or Reader and Data Access)	Log storage account	Read exported logs
Custom read-only Kentik role (optional)	Tenant	Full traffic path views

Test agent in Azure: install a Universal Agent on a small VM with the Synthetics capability, using the same steps as on-prem. It needs outbound TCP 443 only, plus TCP 8877 and UDP 9977 to the other agents.

Source: Kentik KB, Kentik for Azure; Azure Prerequisites & Roles; Flow/Firewall Log Collection (Azure); Kentik Export Configuration for Azure; Automated Configuration (PowerShell)

Google Cloud

No agent. A Log Router sink sends VPC flow logs to Pub/Sub, and Kentik pulls them with its own service account.

Collecting: Topology metadata (required), VPC flow logs. Scope: 2 account(s)/subscription(s)/project(s). Regions: europe-west3.

Steps

1. VPC Network > subnet > Edit > Flow logs On > Configure Logs > Include metadata (Kentik needs it). Any aggregation interval works; sample rate 0 to 100%, default 50%.
2. Logging > Logs Router > Create Sink > Cloud Pub/Sub topic > create the topic, with the inclusion filter below.
3. Pub/Sub > Topics > Create Subscription > delivery type Pull.
4. Pub/Sub > Subscriptions > select it > Add Member > Kentik's service account > Pub/Sub Subscriber and Pub/Sub Viewer.
5. IAM & Admin > IAM > Grant Access > Kentik's service account > Compute Viewer. Whoever grants it needs compute.networks.list and resourcemanager.projects.setIamPolicy.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > GCP Cloud. Enter the project ID, Pub/Sub topic and subscription, name and billing plan, then Save.
7. To cut cost, narrow the sink filter by port, protocol or subnet. A metadata-only export always shows a red flow status; that is expected.

Log Router sink filter

```
resource.type="gce_subnetwork" AND log_id("compute.googleapis.com/vpc_flows")
```

Roles for Kentik's service account (kentik-vpc-flow@kentik-vpc-flow.iam.gserviceaccount.com)

Role	Where	Why
Pub/Sub Subscriber	The pull subscription	Pull flow log messages
Pub/Sub Viewer	The pull subscription	See the subscription
Compute Viewer (not Compute Network Viewer)	Project, or the top-level folder	Read topology metadata

Source: Kentik KB, Kentik for GCP; Cloud Overview; Public Clouds

Oracle Cloud Infrastructure

No agent. OCI Logging captures VCN flow logs, Connector Hub moves them into an Object Storage bucket, and Kentik reads the bucket through OCI APIs.

Collecting: Topology metadata (required), VCN flow logs.

Steps

1. Create an Object Storage bucket for the logs.
2. Create a log group and a capture filter in the root compartment, then turn on VCN flow logs for the VCNs, subnets or VNICs you choose.
3. Create a Connector Hub connector: source Logging (log group and flow log), target Object Storage (bucket, optional object name prefix). Defaults: 100 MB batch, 7 second batch time. The connector can create its own policy.
4. Give Kentik access: the Kentik portal generates a cross-tenancy policy (recommended); paste it into the OCI manual policy editor.

5. In Kentik: Settings > Public Clouds > Create Cloud Export > OCI Cloud. Pick Metadata only, or Flow logs and metadata. Enter tenancy ID, default region (and user ID for a custom user), then Validate Permissions.
6. For flow, enter the bucket name, namespace, connector OCID and prefix. Copy the autogenerated policy, pick a billing plan and Save.
7. Troubleshooting: make sure a custom user is in a group, and that no policy placeholders were left in.

Values to collect for the Kentik export

Value	Where to find it
Tenancy OCID	Tenancy details
User OCID	Custom user only
Home region	The city name in parentheses
Bucket name and namespace	Bucket details
Service connector OCID	Connector details
Object name prefix	Only if you set one
Compartment ID	Defaults to the tenancy OCID

The Traffic Trends and Overviews dashboards are not available for OCI.

Source: Kentik KB, Kentik for OCI; Cloud Overview; Public Clouds

8. Synthetic monitoring plan

Continuous tests measure latency, loss and jitter between sites, clouds and the applications users depend on.

Test points

Agent	Type
Frankfurt DC	Private (Universal Agent)
Amsterdam DC	Private (Universal Agent)
London DC	Private (Universal Agent)
Singapore DC	Private (Universal Agent)
Madrid Branch	Private (Universal Agent)
Milan Branch	Private (Universal Agent)
AWS test agent	Private (Universal Agent)
Azure test agent	Private (Universal Agent)
4 Kentik global agent(s)	Kentik-hosted: Internet hubs, cloud regions, app agents

Planned tests

- Network mesh between all private agents (latency, loss, jitter between every pair of sites and clouds).
- HTTP(S) test to SAP S/4HANA Cloud from private and global agents.
- HTTP(S) test to Microsoft 365 from private and global agents.
- HTTP(S) test to Salesforce from private and global agents.
- HTTP(S) test to Workday from private and global agents.
- BGP Monitor for the customer's prefixes: hijacks, leaks and RPKI status.
- Available test types: Agent-to-Agent, Network Mesh, Server IP, Server Hostname, Network Grid, autonomous tests (ASN, CDN, country, region, city), DNS Server Monitor, DNS Server Grid, HTTP(S) or API, Page Load and Transaction.

Estimated monthly test credits

Test	Interval	Per run	Credits / month
Mesh between 8 private agents	every 5 min	8 x 7 subtests x 2.5 = 140 per run	1,209,600
HTTP(S) tests to 4 SaaS app(s)	every 5 min	8 private + 4 global agents, 5 s timeout = 160 per run	1,382,400
BGP Monitor (2 tests)	continuous	0.53 credits per minute	45,792
Total			2,637,792

Credit rules from the KB: network subtest 2.5 credits from a private agent and 5 from a global agent; HTTP subtests cost timeout (seconds) x 0.5 private or x 1 global; BGP Monitor 0.53 credits per minute; a test every minute runs 43,200 times in a 30-day month; Test Preview is free until saved. Mesh estimates count each direction as one subtest (the KB example: a 4-agent private mesh is 30 credits per run). Treat these as planning estimates and confirm with Kentik.

Source: Kentik KB, Synthetics Overview; Test Types Reference; Test Credits; Agents Page

9. Go-live validation checklist

☐	Frankfurt DC: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Frankfurt DC: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Frankfurt DC: devices discovered in NMS with interfaces and health metrics
☐	Frankfurt DC: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events)
☐	Frankfurt DC: first config backup fetched over SSH
☐	Frankfurt DC: BGP sessions established with Kentik; flows show AS path
☐	Amsterdam DC: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Amsterdam DC: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Amsterdam DC: devices discovered in NMS with interfaces and health metrics
☐	Amsterdam DC: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events)
☐	Amsterdam DC: first config backup fetched over SSH
☐	Amsterdam DC: BGP sessions established with Kentik; flows show AS path
☐	London DC: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	London DC: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	London DC: devices discovered in NMS with interfaces and health metrics
☐	London DC: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events)
☐	London DC: BGP sessions established with Kentik; flows show AS path
☐	Singapore DC: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Singapore DC: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Singapore DC: devices discovered in NMS with interfaces and health metrics
☐	Singapore DC: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events)
☐	Madrid Branch: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Madrid Branch: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Madrid Branch: devices discovered in NMS with interfaces and health metrics
☐	Milan Branch: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Milan Branch: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Milan Branch: devices discovered in NMS with interfaces and health metrics
☐	AWS: cloud export status OK; metadata visible in Kentik Map; flow logs arriving

☐	Azure: cloud export status OK; metadata visible in Kentik Map; flow logs arriving
☐	Google Cloud: cloud export status OK; metadata visible in Kentik Map; flow logs arriving
☐	Oracle Cloud: cloud export status OK; metadata visible in Kentik Map; flow logs arriving
☐	Synthetics: tests show Healthy, Warning or Critical (not Pending); agents Active
☐	Alerting: at least one alert policy routed to the customer's notification channel

Source: Kentik KB, UA Portal Management; Flow Overview; SNMP Traps & Syslog; Public Clouds

10. Security, data handling and licensing

Area	Detail
Hosting	US: Ashburn, VA. EU: Frankfurt. Chosen when the contract is signed.
In transit	TLS 1.2+ to and from Kentik
At rest	AES-256, consistent with FIPS 140-2
Isolation	Separate logical database and tables per customer
Audits	SOC 2 Type II yearly, annual third-party penetration test
Access	SAML 2.0 SSO, MFA (TOTP or hardware keys), user, admin and super admin roles, per-user API tokens
Data	IP addresses and network metadata, never packet payloads; IP redaction available
Retention (default)	45 days full-resolution flow, 120 days summarized, 3 or 13 months NMS metrics by plan; contracts can change this

Source: Kentik KB, Security Overview; KDE Overview

Licensing

License	Covers
FlowPak	Flow from routers, switches, firewalls and hosts, including SNMP flow enrichment.
NMS / DevicePak	Full device monitoring, plus syslog, SNMP traps and SSH config and commands.
CloudPak	Public cloud flow logs, sized in flows per second.
Synthetics	Monthly test credits: global agent subtest 5 credits, private agent 2.5, BGP Monitor 0.53 per minute.

Plans and editions vary by contract. Confirm entitlements with the Kentik account team.

Source: Kentik KB, Licenses; Test Credits

11. Open items to confirm with Kentik

- Which continuous synthetic test types run on the Universal Agent today (the KB names the capability Connectivity Test).
- Licensing names and entitlements per contract (FlowPak, NMS/DevicePak, CloudPak, Synthetics credits).
- Disk sizing for Universal Agent hosts is not published in the KB.
- BGP Proxy shows in Installed Capabilities in the portal, but the KB page still lists it as coming soon. Confirm it is available for this account before relying on it.
- Flow Proxy Forwarding (ktraffic) is marked beta in the KB. Confirm the mode (Forwarding or Edge Processing) per site. Above the KB flow figures, sizing is scaled linearly and is an estimate.

12. Appendix: Kentik KB references

Page	URL
Universal Agents	https://kb.kentik.com/docs/universal-agents
Public Clouds	https://kb.kentik.com/docs/public-clouds
Synthetics Overview	https://kb.kentik.com/docs/synthetics-overview
UA Deployment	https://kb.kentik.com/docs/ua-deployment
NMS Setup	https://kb.kentik.com/docs/nms-setup
SSH Command Access for Kentik NMS	https://kb.kentik.com/docs/ssh-command-access-for-kentik-nms
Kentik for AWS	https://kb.kentik.com/docs/kentik-for-aws
Kentik for Azure	https://kb.kentik.com/docs/kentik-for-azure
Kentik for GCP	https://kb.kentik.com/docs/kentik-for-gcp
Kentik for OCI	https://kb.kentik.com/docs/kentik-for-oci
Test Types Reference	https://kb.kentik.com/docs/test-types
Azure Prerequisites & Roles	https://kb.kentik.com/docs/azure-prerequisites-roles
UA Portal Management	https://kb.kentik.com/docs/ua-portal-management
Device Settings	https://kb.kentik.com/docs/device-settings
Router Configuration	https://kb.kentik.com/docs/router-configuration
Flow Overview	https://kb.kentik.com/docs/flow-overview
Migration Scenarios (kproxy to UA)	https://kb.kentik.com/docs/kproxy-migration-scenarios
SNMP Traps & Syslog	https://kb.kentik.com/docs/snmp-traps-syslogs
Test Credits	https://kb.kentik.com/docs/test-credits
Metadata Configuration (AWS)	https://kb.kentik.com/docs/metadata-configuration-aws
Flow/Firewall Log Collection Configuration (AWS)	https://kb.kentik.com/docs/flow-firewall-log-collection-configuration
Kentik Export Configuration for AWS	https://kb.kentik.com/docs/kentik-export-configuration-aws
Flow/Firewall Log Collection (Azure)	https://kb.kentik.com/docs/flow-firewall-log-collection
Kentik Export Configuration for Azure	https://kb.kentik.com/docs/kentik-export-configuration-azure
Automated Configuration (PowerShell)	https://kb.kentik.com/docs/automated-configuration-powershell
Cloud Overview	https://kb.kentik.com/docs/cloud-overview
Agents Page	https://kb.kentik.com/docs/agents-page
Security Overview	https://kb.kentik.com/docs/security-overview
KDE Overview	https://kb.kentik.com/docs/kde-overview
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